

Sections 2.1 - 2.2

(1)

Geometric Vectors

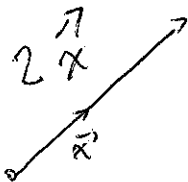
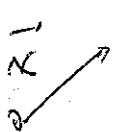
object with direction and magnitude

Vector addition: $\vec{z} = \vec{x} + \vec{y}$



Scalar Multiplication: $\vec{z} = \lambda \vec{x}$

$|\lambda|$ times $\vec{x}$



- direction is the same if $\lambda > 0$

- direction changes if $\lambda < 0$ (opposite)

Vector Subtraction

$$\vec{x} - \vec{y} = \vec{x} + (-\vec{y})$$

Vectors $\Rightarrow$ any object that can be added to another of the same kind and produce a third one of the same kind, and can be multiplied by a scalar to produce an object of the same kind.

Kid.

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Examples:

- polynomials
- elements of $\mathbb{R}^n$.

Systems of linear Equation

$$a_{11}x_1 + \dots + a_{1n}x_n = b_1$$

$$a_{21}x_1 + \dots + a_{2n}x_n = b_2$$

$$\vdots \quad \quad \quad \vdots \quad \quad \quad \vdots$$

$$a_{m1}x_1 + \dots + a_{mn}x_n = b_m$$

Ex 1:

$$x_1 + x_2 + x_3 = 3$$

$$x_1 - x_2 + 2x_3 = 2$$

$$2x_1 + 3x_3 = 1$$

Does it have a solution?
Add (1) + (2)!

No!

What does it mean
to have a solution?

Ex 2:

$$x_1 + x_2 + x_3 = 3$$

$$x_1 - x_2 + 2x_3 = 2$$

$$x_2 + x_3 = 2$$

Same question

$$\begin{aligned} (1) + (2) &\Rightarrow 2x_1 + 3x_3 = 5 \\ (3) - (1) &\Rightarrow x_1 = 1 \end{aligned} \quad \left. \vphantom{\begin{aligned} (1) + (2) &\Rightarrow 2x_1 + 3x_3 = 5 \\ (3) - (1) &\Rightarrow x_1 = 1 \end{aligned}} \right\} x_2 = 1$$

$$x_2 = 1 \quad (1, 1, 1) \rightarrow \text{solution}$$

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Ex 3:

$$x_1 + x_2 + x_3 = 3$$

$$x_1 - x_2 + 2x_3 = 2$$

$$2x_1 + 3x_3 = 5$$

$$(1) + (2) \Rightarrow 2x_1 + 3x_3 = 5$$

$$\Downarrow$$

same as 3rd eq.

$$\text{Take } x_3 = a$$

$$x_1 + x_2 = 3 - a$$

$$x_1 - x_2 = 2 - 2a$$

$$x_1 = \frac{5 - 3a}{2}$$

$$x_2 = \frac{1 + a}{2}$$

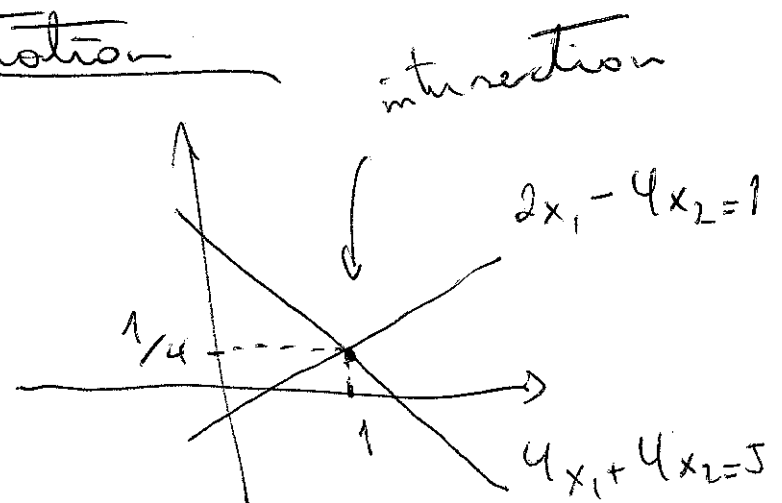
Solution $\left(\frac{5}{2} - \frac{3}{2}a, \frac{1}{2} + \frac{1}{2}a, a\right)$, for $a \in \mathbb{R}$

infinite solutions

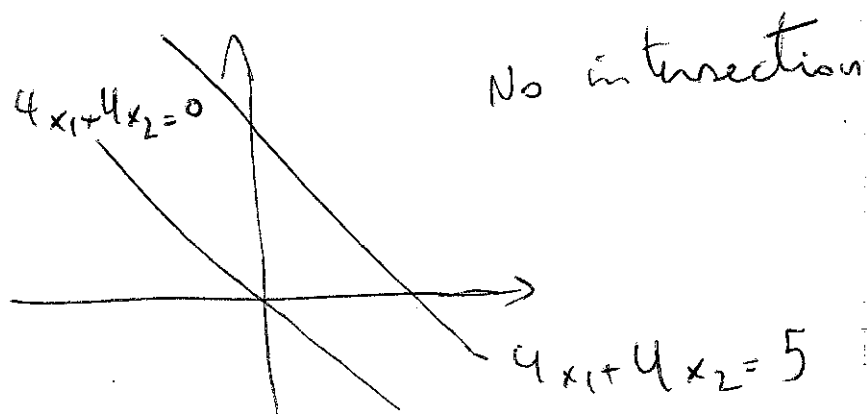
Geometric Interpretation

$$\text{Ex: } \begin{cases} 4x_1 + 4x_2 = 5 \\ 2x_1 - 4x_2 = 1 \end{cases}$$

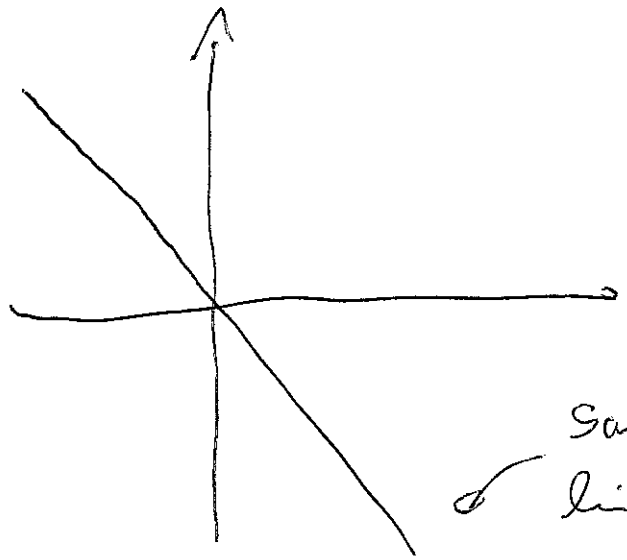
Solution $(1, \frac{1}{4})$



$$\begin{cases} 4x_1 + 4x_2 = 5 \\ 4x_1 + 4x_2 = 0 \end{cases}$$



$$\begin{cases} 4x_1 + 4x_2 = 0 \\ x_1 + x_2 = 0 \end{cases}$$



✓ Same line

inf. many solutions

The system can be rewritten as:

$$\begin{aligned} a_{11}x_1 + \dots + a_{1n}x_n &= b_1 \\ a_{21}x_1 + \dots + a_{2n}x_n &= b_2 \\ &\vdots \\ a_{m1}x_1 + \dots + a_{mn}x_n &= b_m \end{aligned}$$

$$\begin{bmatrix} a_{11} \\ a_{21} \\ \vdots \\ a_{m1} \end{bmatrix} x_1 + \dots + \begin{bmatrix} a_{1n} \\ a_{2n} \\ \vdots \\ a_{mn} \end{bmatrix} x_n = \begin{bmatrix} b_1 \\ b_2 \\ \vdots \\ b_m \end{bmatrix} \quad \text{linear comb of vectors}$$

$$\begin{bmatrix} a_{11} & \dots & a_{1n} \\ a_{21} & \dots & a_{2n} \\ \vdots & & \vdots \\ a_{m1} & \dots & a_{mn} \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix} = \begin{bmatrix} b_1 \\ b_2 \\ \vdots \\ b_m \end{bmatrix} \quad \text{Matrix form}$$

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Def. of matrix

For natural numbers, m, n a (m, n) -matrix A is an $m \times n$ -tuple of elements a_{ij} , $1 \leq i \leq m$, $1 \leq j \leq n$. We write $A \in \mathbb{R}^{m \times n}$ (m rows, n columns).

$$A = \begin{bmatrix} a_{11} & \dots & a_{1n} \\ a_{21} & \dots & a_{2n} \\ \vdots & \ddots & \vdots \\ a_{m1} & \dots & a_{mn} \end{bmatrix}$$

$$B = \begin{bmatrix} b_{11} & \dots & b_{1n} \\ b_{21} & \dots & b_{2n} \\ \vdots & \ddots & \vdots \\ b_{m1} & \dots & b_{mn} \end{bmatrix}$$

Operation

1. Addition: if $A, B \in \mathbb{R}^{m \times n}$, then

$$A + B = \begin{bmatrix} a_{11} + b_{11} & \dots & a_{1n} + b_{1n} \\ \vdots & \ddots & \vdots \\ a_{m1} + b_{m1} & \dots & a_{mn} + b_{mn} \end{bmatrix}$$

2. Scalar Multiplication: if $A \in \mathbb{R}^{m \times n}$, $c \in \mathbb{R}$, then

$$cA = \begin{bmatrix} ca_{11} & \dots & ca_{1n} \\ \vdots & \ddots & \vdots \\ ca_{m1} & \dots & ca_{mn} \end{bmatrix}$$

c. Matrix Mult. if $A \in \mathbb{R}^{m \times n}$ and $B \in \mathbb{R}^{n \times k}$ then

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$$C = AB \in \mathbb{R}^{m \times k}$$

$$C_{ij} = \sum_{l=1}^n a_{il} b_{lj}$$

Ex: $Ax=b \rightarrow A \in \mathbb{R}^{n \times n} \quad x \in \mathbb{R}^{n \times 1} \quad b \in \mathbb{R}^{n \times 1}$

$$b_1 = \sum_{l=1}^n a_{1l} x_l$$

$\uparrow$
 x_l

Remark:

$A \cdot B$ is only defined when # of cols of A

||

of rows of B

$A \cdot B$ might be defined and $B \cdot A$ is not.

• $A \cdot B \neq B \cdot A$

Ex: $A \in \mathbb{R}^{2 \times 3} \quad B \in \mathbb{R}^{3 \times 2} \rightarrow A \cdot B \in \mathbb{R}^{2 \times 2}$
 $B \cdot A \in \mathbb{R}^{3 \times 3}$

Def: $I_n = \begin{bmatrix} 1 & & \\ & \ddots & \\ & & 1 \end{bmatrix} \in \mathbb{R}^{n \times n}$ is called identity matrix
 diag $\rightarrow 1$
 off-diag $\rightarrow 0$.

Properties of matrix product

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- associative

$$(AB)C = A(BC)$$

- distributivity:

$$(A+B)C = AC + BC$$

$$A(C+D) = AC + AD$$

- $\forall A \in \mathbb{R}^{m \times n}$ $I_m A = A$ $A I_n = A$

Def. of Matrix Inverse

For any square matrix $A \in \mathbb{R}^{n \times n}$, if $B \in \mathbb{R}^{n \times n}$ satisfies $A \cdot B = B \cdot A = I_n$, then B is called the inverse of A . $B = A^{-1}$.

$$A = \begin{bmatrix} 1 & 0 \\ 1 & 1 \end{bmatrix} \quad A^{-1} = \begin{bmatrix} 1 & 0 \\ -1 & 1 \end{bmatrix}$$

Note that # of cols = # of rows.

Def. of Matrix Transpose

For $A \in \mathbb{R}^{m \times n}$, the matrix $B \in \mathbb{R}^{n \times m}$ is called the transpose of A ($B = A^T$) if $b_{ij} = a_{ji}$.

$$\text{Ex } A = \begin{bmatrix} 1 & 2 \\ 3 & 4 \\ 5 & 6 \end{bmatrix}$$

$$B = \begin{bmatrix} 1 & 3 & 5 \\ 2 & 4 & 6 \end{bmatrix}$$